

International Institute of Information Technology, Hyderabad

(Deemed to be University)

Answer Booklet for **Mid Sem: Introduction to IoT; Marks: 30, Time: 90 minutes**

Roll No: _____

Room no: _____ Seat: _____

Marks Table (To be filled by the Examiner)

Question No	Marks			Name of the Examiner who marked
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				

Note: For MCQ questions (Q1–Q5), marks are awarded only if ALL correct options are selected. No partial credit.

1. Which of the following statements correctly describe why microcontrollers (MCUs) are preferred over microprocessors (MPUs) in IoT applications? Select exactly THREE.

- (A) MCUs integrate CPU, memory, and peripherals such as timers and ADCs on a single chip
- (B) MCUs offer higher raw processing power than MPUs, making them suitable for heavy data processing
- (C) MCUs are more cost-effective because they do not require external RAM or peripheral interfaces
- (D) MCUs consume less power, making them ideal for battery-operated IoT devices
- (E) MCUs can natively run full operating systems like Linux, giving them more flexibility than MPUs

[1 Mark]

2. Which of the following are true about the ESP32 microcontroller? Select exactly THREE.

- (A) It uses a dual-core Xtensa LX6 processor running at 240 MHz
- (B) It has 520 KB of Flash memory for storing firmware
- (C) It supports both Wi-Fi (802.11 b/g/n) and Bluetooth (Classic + BLE)
- (D) Its default operating system in the ESP-IDF SDK is FreeRTOS
- (E) It has 34 GPIO pins but lacks built-in support for ADC or PWM

[1 Mark]

3. Which of the following are true about the compilation process for ESP32 development? Select exactly TWO.

- (A) Compiling an Arduino sketch on a PC for the ESP32 is an example of cross-compilation because the host (x86) and target (Xtensa) have different architectures
- (B) The Arduino IDE uses Xtensa GCC as the toolchain when compiling for the ESP32
- (C) In native compilation, the compiled binary must still be uploaded to a different platform for execution
- (D) Cross-compilation does not require any special toolchain — the same compiler works for all target architectures
- (E) During the Arduino compilation flow, the sketch (.ino) is directly compiled to a binary without any preprocessing step

[1 Mark]

4. Which of the following statements about SoCs, MCUs, and PCBs are correct? Select exactly THREE.

- (A) A microcontroller like the ESP32 can be considered a basic form of SoC because it integrates CPU, memory, and peripherals on a single chip
- (B) The ESP32-WROOM is a pre-manufactured PCB module that contains the ESP32 chip along with flash memory, antenna, and power regulation circuitry
- (C) Moving from a breadboard prototype to a custom PCB improves signal integrity, reliability, and enables mass production
- (D) A PCB and a breadboard offer the same level of durability and noise performance, so PCBs are only needed for reducing physical size

[1 Mark]

5. Which of the following are true about bare-metal programming and FreeRTOS on the ESP32? Select exactly TWO.

- (A) Bare-metal programming requires the developer to manually handle interrupts, timers, and implement custom multitasking
- (B) FreeRTOS is an optional add-on that must be separately downloaded and linked when compiling with ESP-IDF
- (C) FreeRTOS provides built-in support for multitasking, real-time scheduling, and APIs for memory and task synchronization
- (D) Bare-metal programming on the ESP32 provides better multitasking support than FreeRTOS because it has direct access to hardware
- (E) When you compile an ESP32 program using ESP-IDF, the compiled binary includes only your application code and nothing else

[1 Mark]

6. Differentiate between light sleep and deep sleep in ESP32 with respect to the following:

- (a) Program flow after wake-up
- (b) RAM state
- (c) Time to resume CPU

[3 Marks — 1 each]

7. Consider the following two approaches for reading a temperature sensor every 500ms and a motion sensor every 100ms on an ESP32.

Approach A:

```
unsigned long lastTemp = 0;
unsigned long lastMotion = 0;

void loop() {
    unsigned long now = millis();

    if (now - lastTemp >= 500) {
        readTemperature();
        lastTemp = now;
    }

    if (now - lastMotion >= 100) {
        readMotionSensor();
        lastMotion = now;
    }
}
```

Approach B:

```
void temp_task(void *arg) {
    while (1) {
        readTemperature();
        vTaskDelay(pdMS_TO_TICKS(500));
    }
}

void motion_task(void *arg) {
    while (1) {
        readMotionSensor();
        vTaskDelay(pdMS_TO_TICKS(100));
    }
}
```

```
}  
}
```

Which of the following statements are correct? Select exactly THREE. One mark for one correct choice.

- (A) Approach A is more CPU-efficient than Approach B because `millis()` consumes no processing time
- (B) In Approach B, `vTaskDelay()` suspends the calling task, freeing the CPU for other tasks or idle
- (C) Approach A scales better than Approach B when more sensors are added, as all timing logic stays in one function
- (D) Both approaches achieve approximately correct sampling intervals for the two sensors
- (E) In Approach A, the CPU continuously runs `loop()` checking timestamps, even when no sensor needs reading

[3 Marks]

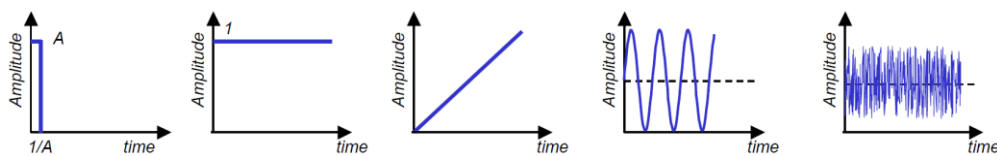
8. A warehouse uses an ESP32 to count boxes passing on a conveyor belt. An optical sensor outputs a HIGH pulse each time a box passes. Boxes can pass quickly, with pulses as short as 20ms. An accurate count is critical for inventory tracking.

- (a) A student uses polling with a 50ms interval to read the optical sensor. Explain why some boxes may go uncounted.
- (b) The student switches to an interrupt-based approach using the ISR + Queue pattern. Explain the role of the ISR and the handler task in this design, and why the processing is split across the two.
- (c) Name two operations that are NOT allowed inside an ISR on the ESP32.

[4 Marks — 1 + 2 + 1]

Sensors & IoT

9. You are testing a sensor with the following input waveforms. Why? (**marks: 1**)



- Sensor response to a variable input is different from what is seen when the input signals are constant (described by static characteristics)
- Dynamic Characteristics are determined by analyzing the response of the sensor to a family of variable input waveforms

10. In what situations would you use an absolute sensor and when would you use a relative sensor? (Use thermocouple and thermistor as examples in your explanation.) (**mark: 2**)

Check if the difference is understood:

Absolute: Detects a stimulus in reference to an absolute physical scale that is independent of the measurement conditions

Relative: Stimulus is sensed with respect to a fixed or variable reference that is not an absolute value independent of measurement conditions

Then check if they have used the thermocouple & thermistor (absolute) in appropriate situations

11. You repeatedly measure the same quantity using a sensor. The readings are very close to each other, but they differ from the true or standard value.

- (a) What type of error does this represent?
- (b) What could be the possible reason (only 1 reason) for this error?

(Marks: 2)

(a) Low accuracy high precision

(b) The reason can be bad calibration or reference value

12. You are part of a team of engineers assigned to design an IoT-based system for forest monitoring. The system must address the following tasks: (Marks: 10)

- a) Detect forest fires
- b) Support ecosystem management (monitor water bodies and water levels, and track tree cover)

Consider two different types of forests:

- i) Dense forests with tall trees and hilly terrain
- ii) Flat forests with short bushes

For each forest type (i and ii) and for each task (a and b), answer the following:

1. List all the data that needs to be collected.
2. Identify the types of sensors (for example: Ultrasound) that could be used to collect this data.
3. Describe at least two challenges of implementing an IoT system in each type of forest.

(Note: marks are not based on ‘how many sentences you write’ but on whether the answer conveys the thought process. So please do not write long answers).

Here, check for the following:

1. The sensors selected were based on the type of forest.
2. Ensure that the data required for making decisions is fairly accurate
3. If they have written limitations – consider it even if any of the above required answer is not very accurate

